

Macroeconomics #1

Some Mathematical Preliminaries

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CASED Pre-Master Macroeconomics
April 23, 2026

Today's lecture

- ① Static Convex Optimization
- ② Dynamic Optimization
- ③ Optimal Control
- ④ Differential Equations
- ⑤ Appendix

Review of Static Convex Optimization

- Objective function: $f(x)$
- Decision vector: $x \in \mathbb{R}^n$.
- Equality constraints: $g_j(x) = c_j, j = 1, \dots, k$, with $n > k$.

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_j(x) = c_j, j = 1, \dots, k$$

Lagrangian:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^k \lambda_j (c_j - g_j(x))$$

First-order conditions:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0, \quad g_j(x^*) = c_j \quad \forall j$$

When constraints are convex, you should also check the SOC, which I believe was covered in Mathematics module.

Test yourself

Find the optimal values of s_t, n_t such that

$$\max_{s_t, n_t} \ln(w_t(1 - \phi n_t) - s_t) + \beta \ln(R_{t+1}s_t) + \gamma \ln(n_t)$$

where:

- n is fertility decision
- ϕ is the time-cost of raising children
- s, w, r are savings, wage rate, and interest rate.

Find the FOC. Verify it is indeed the maximum via SOC.

$$\frac{\partial U_t}{\partial s_t} = -\frac{1}{w_t(1 - \phi n_t) - s_t} + \frac{\beta}{s_t} = 0,$$
$$\frac{\partial U_t}{\partial n_t} = -\frac{\phi w_t}{w_t(1 - \phi n_t) - s_t} + \frac{\gamma}{n_t} = 0,$$

which implies that

$$(s_t) : s_t = \beta c_t,$$
$$(n_t) : n_t = \frac{\gamma c_t}{\phi w_t}$$

Using the budget constraint, we can derive

$$s_t^* = \frac{\beta}{1 + \beta + \gamma} w_t,$$
$$n_t^* = \frac{\gamma}{\phi(1 + \beta + \gamma)}.$$

The Hessian

$$H = \begin{pmatrix} -\frac{1}{\Gamma^2} - \frac{\beta}{s_t^2} & -\frac{\phi w_t}{\Gamma^2} \\ -\frac{\phi w_t}{\Gamma^2} & -\frac{\phi^2 w_t^2}{\Gamma^2} - \frac{\gamma}{n_t^2} \end{pmatrix}$$

where $\Gamma = w_t(1 - \phi n_t) - s_t$. The leading principal minors are

$$|H_1| = -\frac{1}{\Gamma^2} - \frac{\beta}{s_t^2} < 0,$$

$$|H_2| = \begin{vmatrix} -\frac{1}{\Gamma^2} - \frac{\beta}{s_t^2} & -\frac{\phi w_t}{\Gamma^2} \\ -\frac{\phi w_t}{\Gamma^2} & -\frac{\phi^2 w_t^2}{\Gamma^2} - \frac{\gamma}{n_t^2} \end{vmatrix} = \frac{\gamma}{\Gamma^2 n_t^2} + \frac{\beta}{s_t^2} \left(\frac{\phi^2 w_t^2}{\Gamma^2} + \frac{\gamma}{n_t^2} \right) > 0$$

so the solutions obtained at the FOCs are sufficient.

Dynamic Optimization

Infinite horizon planner: cake-eating problem

Initial cake stock $k_0 > 0$, no restock:

$$k_{t+1} = k_t - c_t, \quad c_t \geq 0, \quad k_{t+1} \geq 0.$$

Planner chooses $\{c_t\}_{t \geq 0}$ to solve

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad 0 < \beta < 1.$$

- State variable: k_t (the remaining cake).
- Control variable: c_t (current consumption slice).

(*) To be rigorous, the max operator should be replaced by sup as we cannot guarantee that the optimum exists. For simplicity, let's assume the problem satisfies Blackwell's Sufficient conditions for a max to exist and be unique.

Lagrangian approach

The Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (k_t - c_t - k_{t+1})]$$

FOC:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} &= \beta^t [u'(c_t) - \lambda_t] = 0 \iff \lambda_t = u'(c_t), \\ \frac{\partial \mathcal{L}}{\partial k_{t+1}} &= -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} = 0 \iff \lambda_t = \beta \lambda_{t+1} \end{aligned}$$

Substitute $\lambda = u'(c_t)$ to obtain the Euler equation

$$u'(c_t) = \beta u'(c_{t+1})$$

The solution is a sequence of consumption path $\{c_t\}_{t=0}^{\infty}$.

Recursive Structure

By define the value function given an initial cake size k

$$V(k) = \max \sum_{t=0}^{\infty} \beta^t u(c) \quad (1)$$

We obtain maximum lifetime utility starting with cake. Assuming that all future optimal values have been found, the objective function can be written as

$$\sum_{t=0}^{\infty} \beta^t u(c_t) = u(c_0) + \beta \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \quad (2)$$

or (suppressing the time notation)

$$V(k) = u(c) + \beta V(k - c) \quad (3)$$

This exploits the principle of optimality: An optimal plan from today must remain optimal tomorrow and any future point onward.

Bellman formulation of cake-eating

The Bellman equation:

$$V(k) = \max_{0 \leq c \leq k} \{u(c) + \beta V(k - c)\}.$$

Equivalent choice in next-state form:

$$V(k) = \max_{0 \leq k' \leq k} \{u(k - k') + \beta V(k')\}.$$

This is the canonical one-state dynamic programming model.

- state variable: k
- choice variable: k'

FOC and envelope for interior solution

FOC:

$$u'(c) = \beta V'(k').$$

Envelope condition:

$$V'(k) = u'(k - k') = u'(c)$$

Shift envelope one period:

$$V'(k') = u'(c').$$

So Euler equation becomes

$$u'(c) = \beta u'(c').$$

Euler equation

The same Euler equation is obtained

$$u'(c_t) = \beta u'(c_{t+1})$$

Intuition:

- LHS: marginal utility **loss** from reducing c_t by one unit
- RHS: discounted marginal utility **gain** from adding that unit to tomorrow.

The agent obtain optimal consumption at the point where LHS=RHS.
If u is concave, planner smooths consumption over time.

Closed-form solution for log utility

Let $u(c) = \log c$. Euler:

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \Rightarrow c_{t+1} = \beta c_t.$$

So consumption is geometric:

$$c_t = \beta^t c_0.$$

Resource feasibility requires

$$\sum_{t=0}^{\infty} c_t = k_0$$

This implies

$$c_0(1 + \beta + \beta^2 + \cdots) = k_0 \quad (4)$$

Applying geometric sequence

$$c_0 = (1 - \beta)k_0 \quad (5)$$

Policy and value functions for log utility

The policy function is a function taking the state variable as input to output the choice/control variable

$$c_t = h(k_t) \tag{6}$$

Optimal policy:

$$c_t^* = (1 - \beta)k_t \quad k_{t+1}^* = \beta k_t.$$

We can also solve for the Value function $V(k)$ by Guess-and-Verify. Assume that the value function takes the form :

$$V(k) = A + B \log k.$$

This guess is reasonable since the objective function is a log function, and that it satisfies Blackwell's Sufficient conditions. One can solve for A, B as [▶ See derivation](#)

$$B = \frac{1}{1 - \beta}, \quad A = \frac{\log(1 - \beta)}{1 - \beta} + \frac{\beta \log \beta}{(1 - \beta)^2}.$$

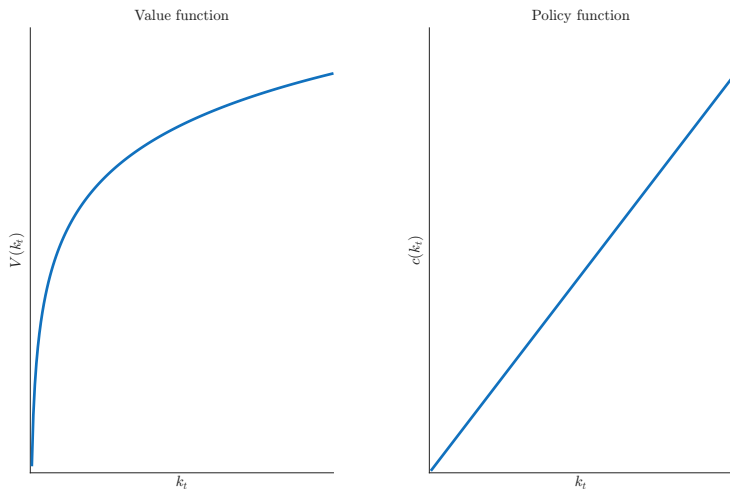


Figure: Value function and Policy function as functions of k_t

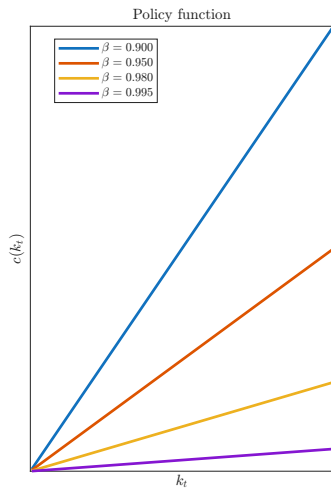
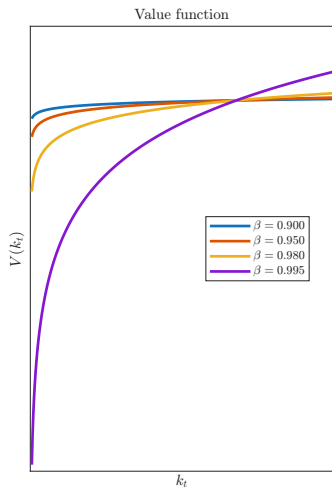
Comparative statics:

$$c_0 = (1 - \beta)k_0, \quad k_{t+1} = \beta k_t.$$

- Higher β (more patient): lower initial consumption, slower cake depletion.
- Lower β : front-loaded consumption and faster depletion.

However, the Euler equation is not sufficient to guarantee optimality. It does not rule out the paths that satisfy the Euler equation but leave valuable cake unconsumed forever. When it happens, we could increase consumption slightly in the future and improve utility, but then, the chosen consumption path will no longer optimal.

Therefore, we need a transversality condition to rule out those paths.



Deriving the transversality condition

FOC:

$$-\beta^t u'(c_t) + \beta^{t+1} u'(c_{t+1}) = 0$$

Multiply by k_{t+1} and sum:

$$\sum_{t=0}^T \left[-\beta^t u'(c_t) k_{t+1} + \beta^{t+1} u'(c_{t+1}) k_{t+1} \right] = 0$$

Split sums and then shift the time index of the second term

$$\begin{aligned} -\sum_{t=0}^T \beta^t u'(c_t) k_{t+1} + \sum_{t=0}^T \beta^{t+1} u'(c_{t+1}) k_{t+1} &= 0 \\ -\sum_{t=0}^T \beta^t u'(c_t) k_{t+1} + \sum_{s=1}^{T+1} \beta^s u'(c_s) k_s &= 0 \quad (s = t + 1) \end{aligned}$$

Everything cancels out except the two endpoints

$$-\beta^0 u'(c_0) k_1 + \beta^{T+1} u'(c_{T+1}) k_{T+1} = 0 \tag{7}$$

Deriving the transversality condition

Rearranging:

$$\beta^0 u'(c_0) k_1 = \beta^{T+1} u'(c_{T+1}) k_{T+1}$$

Taking $T \rightarrow \infty$:

$$\beta^0 u'(c_0) k_1 = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1}$$

For optimality, the RHS must not explode, so we must have:

$$\lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0$$

Interpretation:

- k_{T+1} : leftover cake
- $u'(c_T)$: value of cake

The TVC ensures that **no** valuable resources are left unused at infinity.

Transversality condition in cake-eating

For this model:

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

With log utility and optimal policy:

$$\beta^t \frac{1}{c_t} k_{t+1} = \beta^t \frac{1}{(1 - \beta) \beta^t k_0} (\beta^{t+1} k_0) = \frac{\beta^{t+1}}{1 - \beta}.$$

And

$$\lim_{t \rightarrow \infty} \frac{\beta^{t+1}}{1 - \beta} = 0$$

since $\beta < 1$, the TVC is satisfied.

Exercise: dynamic programming with depreciation

If you do not eat the cake, it gets rotten a bit by a fraction of $\delta \in (0, 1)$. Find the optimal $\{c_t\}_{t \geq 0}$ to solve

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

subject to

$$k_{t+1} = (1 - \delta)k_t - c_t, \quad 0 < \delta < 1,$$

with $k_0 > 0$ and $0 \leq c_t \leq (1 - \delta)k_t$.

- 1 Write the Bellman equation with state k and control c (or k').
- 2 Derive the FOC and envelope condition, then combine them into the Euler equation.
- 3 Guess that the policy is linear, $c_t = \alpha k_t$, and solve for α as a function of β and δ .
- 4 Verify whether the implied law of motion for k_t satisfies the TVC.

Optimal Control

The cake-eating problem in continuous time

Objective function

$$\max_{c(t)} \int_0^{\infty} e^{-\rho t} u(c(t)) dt$$

subject to the dynamics

$$\dot{k}(t) = -c(t)$$

- Initial condition $k(0) = k_0 > 0$
- Control: $c(t) \geq 0$

Some macro models are naturally written in continuous time. You will see a lot in labor economics and macroeconomics. Optimal control is the analog of dynamic programming in continuous time.

From discrete to continuous time

Start from:

$$\sum_{t=0}^{\infty} \beta^t u(c_t), \quad k_{t+1} = k_t - c_t$$

Introduce Δt such that $t = n\Delta t$. Since $0 < \beta < 1$, I can write $\beta = e^{-\rho}$.
Rescale so that stock = flow $\times \Delta t$

$$\sum_{t=0}^{\infty} \beta^t u(c_t) \rightarrow \sum_{n=0}^{\infty} e^{-\rho n \Delta t} u(c(n\Delta t)) \Delta t$$

As $\Delta t \rightarrow 0$, take the limits (Riemann sum)

$$\lim_{\Delta t \rightarrow 0} \sum_{n=0}^{\infty} e^{-\rho n \Delta t} u(c(n\Delta t)) \Delta t = \int_0^{\infty} e^{-\rho t} u(c(t)) dt$$

Law of motion. From $k_{t+1} = k_t - c_t$ or $k(t + \Delta t) = k(t) + c(t)\Delta t$

$$\frac{k(t + \Delta t) - k(t)}{\Delta t} = -c(t)$$

Taking the limits: $\Delta t \rightarrow 0$ we obtain $\dot{k}(t) = -c(t)$

To solve this problem, write the present-value Hamiltonian

$$\mathcal{H} = e^{-\rho t} u(c(t)) + \lambda(t) \dot{k}(t)$$

[*] Note that $\lambda(t)$ is also called the co-state variable. The FOC:

$$(c(t)) : \quad \frac{\partial \mathcal{H}}{\partial c} = e^{-\rho t} u'(c(t)) - \lambda(t) = 0,$$

$$(\dot{k}(t)) : \quad \dot{k}(t) = \frac{\partial \mathcal{H}}{\partial \lambda},$$

$$(\dot{\lambda}(t)) : \quad \dot{\lambda}(t) = -\frac{\partial \mathcal{H}}{\partial k}$$

And the transversality condition

$$\lim_{t \rightarrow \infty} \lambda(t) k(t) = 0$$

In the cake-eating problem, we have:

$$e^{-\rho t} u'(c(t)) = \lambda(t)$$

State equation: $\dot{k}(t) = -c(t)$.

Co-state equation:

$$\dot{\lambda}(t) = 0$$

since $\mathcal{H}$ is independent of k . This implies $\lambda(t) = \lambda_0 \quad \forall t$.

Differentiate $\lambda(t)$ wrt time t , we obtain:

$$\dot{\lambda}(t) = -\rho e^{-\rho t} u'(c(t)) + e^{-\rho t} u''(c(t)) \dot{c}(t)$$

But $\dot{\lambda}(t) = 0$ so

$$-\rho u'(c(t)) + u''(c(t)) \dot{c}(t) = 0$$

Rearranging to obtain the Euler equation

$$u''(c) \dot{c} = \rho u'(c)$$

The transversality condition

$$\lim_{t \rightarrow \infty} \lambda(t) k(t) = e^{-\rho t} u'(c(t)) k(t) = 0.$$

Solution

Assume that $u(c) = \log c$, then the Euler equation becomes

$$\frac{\dot{c}(t)}{c(t)} = -\rho$$

The LHS is actually

$$\frac{\dot{c}(t)}{c(t)} = \frac{d}{dt} \log c(t) = -\rho$$

Integrate both sides wrt t :

$$\int \frac{d}{dt} \ln c(t) dt = \int -\rho dt$$

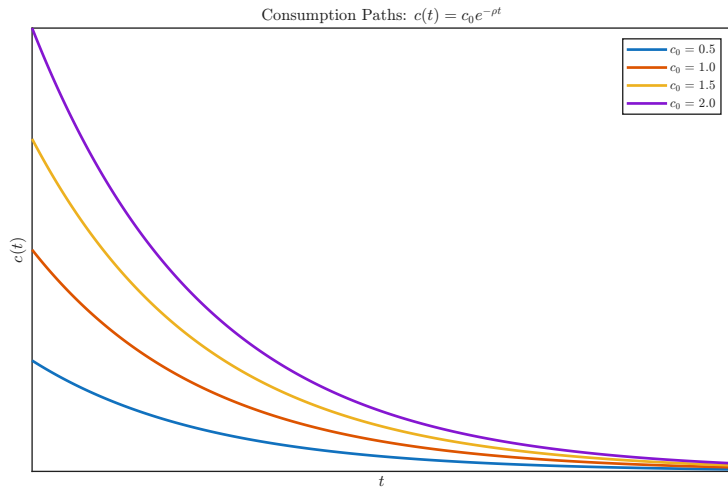
or

$$\ln c(t) = -\rho t + C$$

Exponentiate both sides

$$c(t) = e^{-\rho t + C} = e^{-\rho t} c_0. \quad (c_0 \equiv e^C)$$

Time path



The HJB Equation

The Hamilton–Jacobi–Bellman (HJB) equation is the continuous-time analogue of Bellman ► Bellman to HJB

$$V(k) = \max_{c \geq 0} \{u(c) + \beta V(k')\}$$

In continuous time:

$$\rho V(k) = \max_{c \geq 0} \{u(c) + V'(k)\dot{k}\}$$

In the cake-eating problem ($\dot{k} = -c$):

$$\rho V(k) = \max_{c \geq 0} \{u(c) - cV'(k)\}$$

Interpretation:

- LHS: required return on the value function
- RHS: flow utility + change in value of the state

From HJB to Dynamics of $V'(k)$

HJB:

$$\rho V(k) = \max_c \{u(c) - cV'(k)\}$$

First-order condition:

$$u'(c) = V'(k)$$

Differentiate the HJB with respect to k :

$$\rho V'(k) = \frac{d}{dk} [u(c) - cV'(k)]$$

Using the chain rule:

$$\rho V'(k) = u'(c) \frac{\partial c}{\partial k} - \left[\frac{\partial c}{\partial k} V'(k) + cV''(k) \right]$$

Using $u'(c) = V'(k)$, the first two terms cancel:

$$\rho V'(k) = -cV''(k)$$

Euler Equation from HJB

From the previous slide:

$$\rho V'(k) = -cV''(k)$$

Differentiate wrt time

$$\dot{V}'(k) = V''(k)\dot{k}$$

Since $\dot{k} = -c$:

$$\dot{V}'(k) = -cV''(k)$$

Substitute:

$$\dot{V}'(k) = \rho V'(k)$$

Using the FOC $V'(k) = u'(c)$:

$$\dot{u}'(c) = \rho u'(c)$$

For $u(c) = \log c$, $u'(c) = \frac{1}{c}$:

$$-\frac{1}{c^2}\dot{c} = \rho\frac{1}{c} \Rightarrow \frac{\dot{c}}{c} = -\rho$$

Exercise: optimal control with regeneration

A renewable resource stock evolves according to

$$\dot{k}(t) = ak(t) - c(t), \quad a > 0,$$

and the planner solves

$$\max_{c(t)} \int_0^{\infty} e^{-\rho t} \log c(t) dt$$

subject to $k(0) = k_0 > 0$ and $c(t) \geq 0$.

- 1 Write the present-value Hamiltonian.
- 2 Derive the FOC, state equation, and co-state equation.
- 3 Show that the optimal consumption growth rate satisfies an Euler equation of the form $\dot{c}/c = \text{constant}$, and identify that constant.
- 4 Use the TVC to discuss the restriction on parameters needed for an economically meaningful solution.

Differential Equations

What is an Autonomous ODE?

An **autonomous** ordinary differential equation is of the form

$$\dot{x}(t) = f(x(t))$$

where the right-hand side does not explicitly depend on time t .

- The system's behavior is determined solely by the current state x .
- Examples: $\dot{x} = rx$ (exponential growth), $\dot{x} = x(1 - x)$ (logistic).
- We will start with one variable, then extend to two variables.

Linear Autonomous ODE: $\dot{x} = \lambda x$

Theorem

The solution to $\dot{x} = \lambda x$ with initial condition $x(0) = x_0$ is

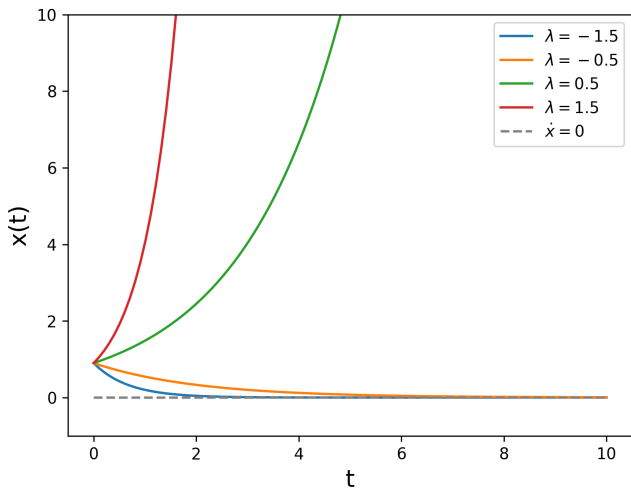
$$x(t) = x_0 e^{\lambda t}.$$

Example

If $\lambda = -2$ and $x_0 = 5$, then $x(t) = 5e^{-2t} \rightarrow 0$ as $t \rightarrow \infty$.

Definition (Stability)

- If $\lambda < 0$: $x(t) \rightarrow 0$ (asymptotically stable).
- If $\lambda > 0$: $x(t) \rightarrow \infty$ (unstable).
- If $\lambda = 0$: $x(t) = x_0$ (neutrally stable).



Inhomogeneous Case: $\dot{x} = \lambda x + b$

Theorem

The general solution is

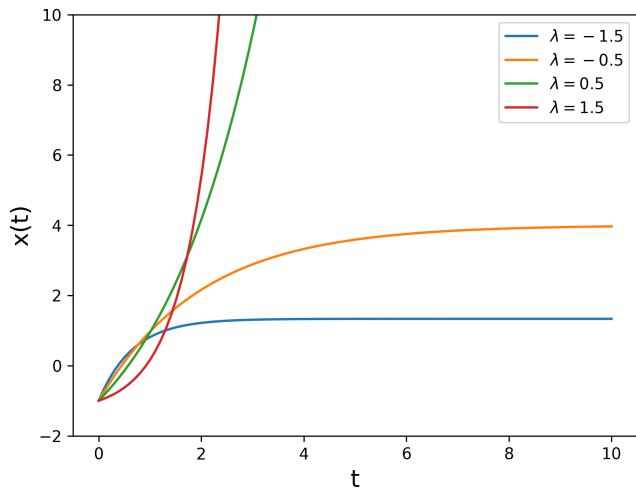
$$x(t) = -\frac{b}{\lambda} + ke^{\lambda t},$$

where $k = x_0 + \frac{b}{\lambda}$.

- Steady state (equilibrium): $\bar{x} = -\frac{b}{\lambda}$.
- Stability depends on sign of λ (same as homogeneous case).

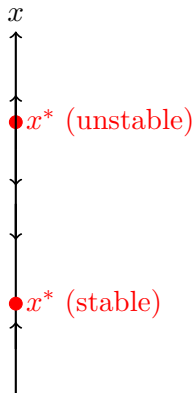
Example

$\dot{x} = -2x + 4$. Then $\bar{x} = 2$, and $x(t) = 2 + (x_0 - 2)e^{-2t}$.



Phase Line for One Variable

For $\dot{x} = f(x)$, the phase line shows direction of motion.



- Fixed points where $f(x) = 0$.
- Arrows indicate sign of $\dot{x}$.

Nonlinear One-Variable: Linearization

Consider $\dot{x} = f(x)$ with equilibrium $\bar{x}$ ($f(\bar{x}) = 0$). Taylor expansion:
 $f(x) \approx f'(\bar{x})(x - \bar{x})$.

Theorem (Stability of equilibria)

- If $f'(\bar{x}) < 0$, then $\bar{x}$ is locally asymptotically stable.
- If $f'(\bar{x}) > 0$, then $\bar{x}$ is unstable.
- If $f'(\bar{x}) = 0$, inconclusive (need higher order).

Example

$\dot{x} = x(1 - x)$. Equilibria: $x = 0$, $x = 1$. $f'(x) = 1 - 2x$. $f'(0) = 1 > 0$ (unstable), $f'(1) = -1 < 0$ (stable).

Nonlinear ODE case: logistic growth

Consider the nonlinear ODE

$$\dot{x} = f(x) = rx \left(1 - \frac{x}{K}\right), \quad r > 0, K > 0.$$

Its steady states solve $\dot{x} = 0$, so $x^* = 0$ and $x^* = K$. To study local stability, compute

$$f'(x) = r - \frac{2r}{K}x.$$

Therefore

$$f'(0) = r > 0 \Rightarrow x^* = 0 \text{ is unstable,}$$

and

$$f'(K) = -r < 0 \Rightarrow x^* = K \text{ is locally asymptotically stable.}$$

- If $0 < x < K$, then $\dot{x} > 0$ so x rises (toward K)
- If $x > K$ (or $x < 0$), then $\dot{x} < 0$ so x falls back toward K (or far from 0).

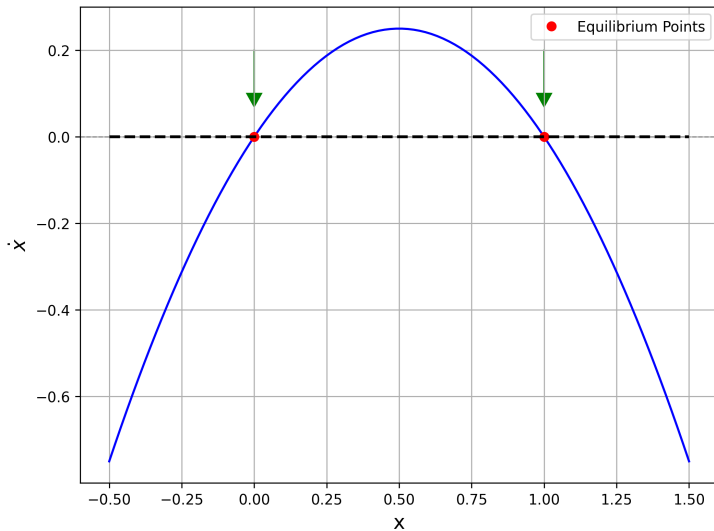
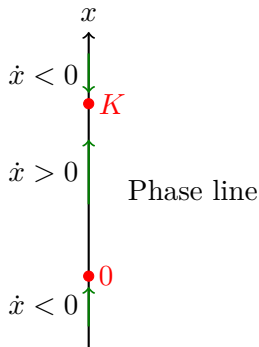


Figure: logistic growth

Logistic growth: phase diagram



For positive initial conditions, trajectories move away from 0 and converge to K .

Linear System of Two ODEs

Consider

$$\begin{cases} \dot{x} = ax + by, \\ \dot{y} = cx + dy. \end{cases}$$

In matrix form: $\dot{\mathbf{x}} = A\mathbf{x}$, with $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Goal

Find explicit solution using eigenvalues and eigenvectors.

Assume A has two distinct real eigenvalues λ_1, λ_2 with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$. Then general solution:

$$\mathbf{x}(t) = C_1 e^{\lambda_1 t} \mathbf{v}_1 + C_2 e^{\lambda_2 t} \mathbf{v}_2.$$

Stability Classification for Linear Systems

The steady state is $(0,0)$.

Theorem

Let λ_1, λ_2 be eigenvalues of A .

- **Sink (stable):** $\lambda_1 < 0, \lambda_2 < 0$ and $|A| > 0$.
- **Source (unstable):** $\lambda_1 > 0, \lambda_2 > 0$ and $|A| > 0$.
- **Saddle:** eigenvalues have opposite signs and $|A| < 0$
- For complex eigenvalues $\alpha \pm i\beta$: stability determined by α (negative $\rightarrow$ stable focus, positive $\rightarrow$ unstable focus).

Example:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}: \text{eigenvalues } -1, -2 \rightarrow \text{sink (stable)}.$$

How to draw a phase diagram

- 1 Solve $\dot{k} = 0$ locus.
- 2 Solve $\dot{c} = 0$ locus.
- 3 Determine signs of $\dot{k}, \dot{c}$ in each region.
- 4 Add motion arrows.
- 5 Identify the stable arm and transition path from k_0 .

Example

Let us consider the system

$$\begin{aligned}\dot{x} &= ax + by + \kappa_1, \\ \dot{y} &= cx + dy + \kappa_2.\end{aligned}\tag{8}$$

where $a < 0, b < 0, c < 0, d > 0, \kappa_1 > 0, \kappa_2 > 0$.

We can compute that

$$|A| = ad - bc < 0$$

so the dynamical system is likely a “saddle”.

Step 1: Solve the nullclines

- The first locus $\dot{x} = 0$ implying $ax + by + \kappa_1 = 0$, so the $\dot{x} = 0$ locus is

$$y_x = -\frac{a}{b}x - \frac{\kappa_1}{b}$$

since $a < 0, b < 0$, the locus is a straight line with a negative slope (wrt x)

- The second locus $\dot{y} = 0$ is given by

$$y_y = -\frac{c}{d}x - \frac{\kappa_2}{d}$$

since $c < 0, d > 0$, the locus is a straight line with a positive slope (wrt x)

Step 2: Solve for the steady state

The steady state is given by the intersection of the two nullclines. Let's call it (x^*, y^*) . They are found by solving

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} -\kappa_1 \\ -\kappa_2 \end{pmatrix}$$

the solution is

$$x^* = \frac{b\kappa_2 - d\kappa_1}{ad - bc},$$
$$y^* = \frac{c\kappa_1 - a\kappa_2}{ad - bc}$$

Together with the nullclines, it divides the (x, y) plane into 4 areas.

We now determine the signs in each region

- Given y , the locus $\dot{x} = 0$ indicates there is no change in x .

$$\dot{x} = ax + by + \kappa_1$$

Use the nullcline to rewrite

$$\dot{x} = b(y - y_x(x))$$

Since $b < 0$:

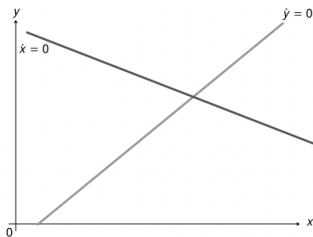
- If $y > y_x(x)$ – above the locus $\dot{x} = 0$, we have $y - y_x > 0$ so $\dot{x} < 0$
- If $y < y_x(x)$ – below the locus $\dot{x} = 0$, we have $y - y_x < 0$ so $\dot{x} > 0$
- Given x , the locus $\dot{y} = 0$ indicates there is no change in y

$$\dot{y} = cx + dy + \kappa_2 = d(y - y_y(x))$$

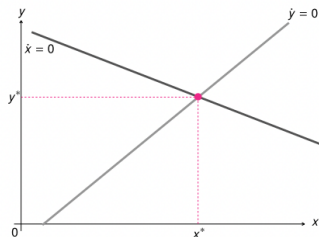
Since $d > 0$:

- If $y > y_y(x)$ – above the locus $\dot{y} = 0$, we have $y - y_y > 0$ so $\dot{y} > 0$
- If $y < y_y(x)$ – below the locus $\dot{y} = 0$, we have $y - y_y < 0$ so $\dot{y} < 0$

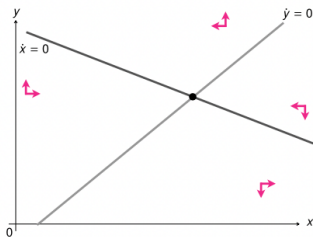
Phase Plane: Nullclines and Trajectories



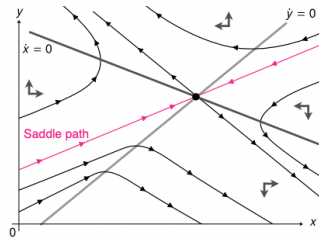
A. Nullclines



B. Steady state



C. Directional arrows



D. Trajectories

Some tricks when determining the signs

- With the nullcline $\dot{x} = 0$, only pay attention to the derivatives of $\dot{x}$ with respect to y . Since $b < 0$, $\dot{x}$ flips sign above/below the line.
- With the nullcline $\dot{y} = 0$, only pay attention to the derivatives of $\dot{y}$ with respect to y . Since $d > 0$, $\dot{y}$ retains the signs above/below the line.

► How to derive the stable arm

Phase Plane: Nullclines and Trajectories

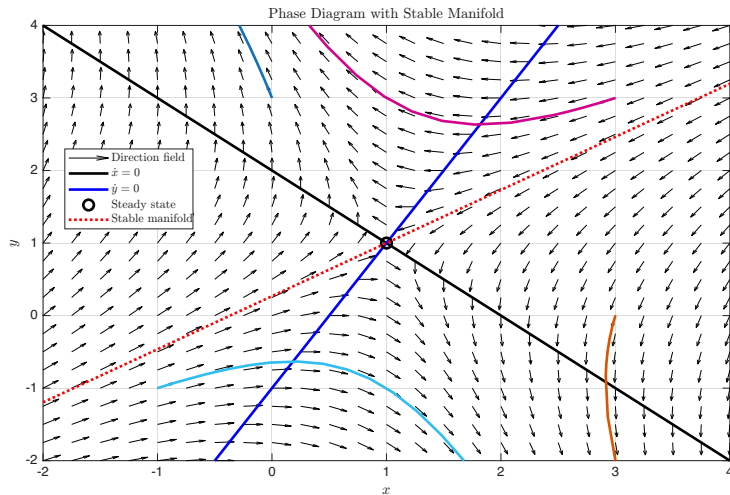


Figure: Linear system

Nonlinear Systems: Linearization

Consider $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ with equilibrium $(\bar{x}, \bar{y})$. Linearize:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} \approx J(\bar{x}, \bar{y}) \begin{pmatrix} x - \bar{x} \\ y - \bar{y} \end{pmatrix},$$

where $J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}$ is the Jacobian.

Stability via Trace and Determinant

Let $\tau = \text{tr}(J)$, $\Delta = \det(J)$.

- If $\Delta > 0$ and $\tau < 0$: stable node or focus.
- If $\Delta > 0$ and $\tau > 0$: unstable node or focus.
- If $\Delta < 0$: saddle point.
- If the roots are complex $\lambda = \alpha \pm i\beta$:
 - $\alpha < 0$: stable focus (spiral in).
 - $\alpha > 0$: unstable focus (spiral out).
 - $\alpha = 0$: center (closed orbits).

Example: Nonlinear System

Consider the following system

$$\dot{x} = -x + y^2$$

$$\dot{y} = -y.$$

- Equilibrium solve $-y = 0 \Rightarrow y = 0$, then $-x + 0 = 0 \Rightarrow x = 0$.
- Stability Jacobian: $J = \begin{pmatrix} -1 & 2y \\ 0 & -1 \end{pmatrix}$.

At $(0,0)$: $J = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. The eigenvalues are: $-1, -1 \rightarrow$ stable node.

General forms:

$$\dot{x} = F(x), \quad x \in \mathbb{R}^m.$$

- Steady state: $F(x^*) = 0$.
- Local dynamics determined by linearization near x^* .

For 2D system, we linearize the system about the steady state

$$\begin{cases} \dot{k} = G(k, c) \\ \dot{c} = H(k, c) \end{cases}$$

around (k^*, c^*) :

$$\begin{bmatrix} \dot{k} \\ \dot{c} \end{bmatrix} \approx J \begin{bmatrix} k - k^* \\ c - c^* \end{bmatrix}, \quad J = \begin{bmatrix} G_k & G_c \\ H_k & H_c \end{bmatrix}_{(k^*, c^*)}.$$

- Two negative eigenvalues: stable node.
- Two positive eigenvalues: unstable node.
- Opposite signs: saddle.
- Complex roots: spiral behavior.

Example 2

Suppose

$$\dot{k} = ak - c, \quad \dot{c} = b(k - \bar{k}),$$

with $a, b > 0$. Steady state:

$$k^* = \bar{k}, \quad c^* = a\bar{k}.$$

Jacobian:

$$J = \begin{bmatrix} a & -1 \\ b & 0 \end{bmatrix}.$$

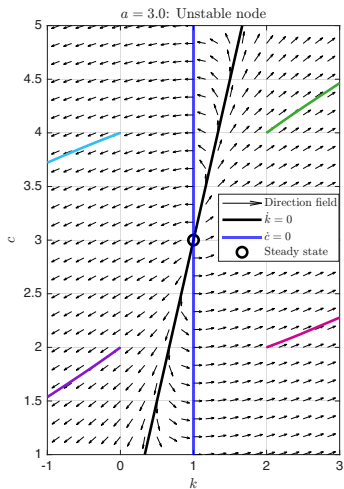
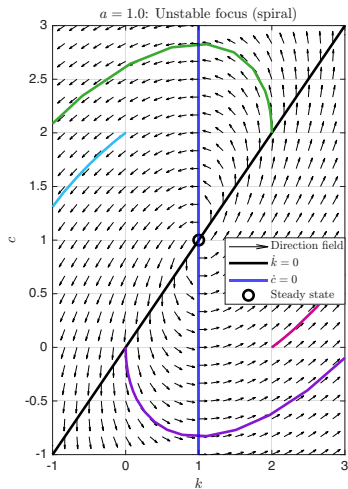
Characteristic equation:

$$\lambda^2 - a\lambda + b = 0.$$

Determinant and trace:

$$\det(J) = b > 0, \quad \text{tr}(J) = a > 0.$$

If discriminant $a^2 - 4b > 0$, both eigenvalues are real and positive for sufficiently large a , in that case we have unstable node; with other parameter sets one can get saddle/spiral.



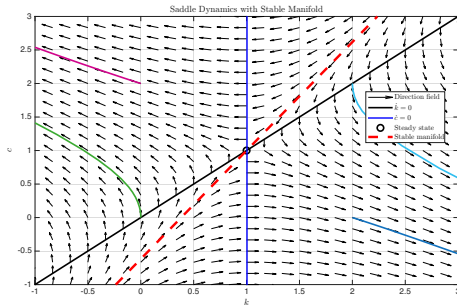
If we modify the previous system by flipping the sign of b

$$\dot{k} = ak - c, \quad \dot{c} = -b(k - \bar{k}),$$

with $a, b > 0$. Then the Jacobian matrix becomes

$$J = \begin{bmatrix} a & -1 \\ -b & 0 \end{bmatrix}.$$

so $\det(J) = -b < 0$, we will have a saddle.



APPENDIX

Guess and Verify

The Bellman becomes

$$V(k) = A + B \log k = \max_c \log c + \beta[A + B \log(k - c)]$$

The FOC

$$\frac{1}{c} = \frac{\beta B}{k - c}$$

implying

$$c^* = k / (1 + \beta B),$$

$$k' = k - c^* = k - \frac{k}{1 + \beta B} = \frac{\beta B}{1 + \beta B} k$$

This means:

$$\log c^* = \log k - \log(1 + \beta B),$$

$$\log(k - c^*) = \log k + \log\left(\frac{\beta B}{1 + \beta B}\right)$$

Substituting all to the Bellman equation

$$V(k) = \log k - \log(1 + \beta B) + \beta A + \beta B[\log k + \log(\beta B / (1 + \beta B))]$$

Matching coefficients

$$(\log k) : B = 1 + \beta B \implies B = 1/(1 - \beta)$$

The constant part

$$A(1 - \beta) = -\log(1 + \beta B) + \beta B \log(\beta B / (1 + \beta B))$$

Plugging in $B = 1/(1 - \beta)$ and solve for A yields

$$A = \frac{\log(1 - \beta)}{1 - \beta} + \frac{\beta \log \beta}{(1 - \beta)^2}$$

Solution manual: DP exercise

Problem:

$$V(k) = \max_{0 \leq c \leq (1-\delta)k} \left\{ \log c + \beta V((1-\delta)k - c) \right\}.$$

FOC and envelope:

$$\frac{1}{c_t} = \beta V'(k_{t+1}), \quad V'(k_t) = \frac{1-\delta}{c_t}.$$

Shifting the envelope one period gives $V'(k_{t+1}) = (1-\delta)/c_{t+1}$, hence

$$\frac{1}{c_t} = \beta \frac{1-\delta}{c_{t+1}} \quad \Rightarrow \quad c_{t+1} = \beta(1-\delta)c_t.$$

Guess $c_t = \alpha k_t$. Then

$$k_{t+1} = (1-\delta-\alpha)k_t, \quad \alpha(1-\delta-\alpha) = \beta(1-\delta)\alpha.$$

For $\alpha > 0$, we get $\alpha = (1-\beta)(1-\delta)$ and thus the policy:

$$c_t = (1-\beta)(1-\delta)k_t, \quad k_{t+1} = \beta(1-\delta)k_t.$$

For the DP exercise, the policy is

$$c_t = (1 - \beta)(1 - \delta)k_t, \quad k_t = [\beta(1 - \delta)]^t k_0.$$

Then

$$\beta^t u'(c_t) k_{t+1} = \beta^t \frac{1}{c_t} k_{t+1} = \frac{[\beta(1 - \delta)]^{t+1}}{(1 - \beta)(1 - \delta)} \rightarrow 0,$$

so the TVC holds because $\beta(1 - \delta) < 1$.

Solution for the Optimal Control

For the optimal control exercise:

$$\mathcal{H} = e^{-\rho t} \log c(t) + \lambda(t)[ak(t) - c(t)].$$

Necessary conditions are

$$e^{-\rho t} \frac{1}{c(t)} = \lambda(t), \quad \dot{k} = ak - c, \quad \dot{\lambda} = -a\lambda.$$

Differentiating $\lambda = e^{-\rho t}/c$ wrt time and equating to $-a\lambda$ gives

$$\frac{\dot{c}(t)}{c(t)} = a - \rho.$$

Thus consumption grows at rate $a - \rho$; for the TVC to be economically meaningful, discounted value must vanish asymptotically, which requires $\rho > a$ so consumption does not grow too fast relative to discounting.

From Discrete Bellman to Continuous Time (1-3)

Start from the discrete-time Bellman equation:

$$V(k_t) = \max_{c_t} \{u(c_t) + \beta V(k_{t+1})\}$$

subject to:

$$k_{t+1} = k_t - c_t$$

Introduce a small time step Δt :

- Flow utility: $u(c_t)\Delta t$
- Discounting: $\beta = e^{-\rho\Delta t}$
- Law of motion: $k_{t+\Delta t} = k_t - c_t\Delta t$

Rewrite Bellman equation:

$$V(k) = \max_c \left\{ u(c)\Delta t + e^{-\rho\Delta t} V(k - c\Delta t) \right\}$$

Taylor Expansion and Rearrangement (4–7)

Use first-order approximations:

$$\begin{aligned}V(k - c\Delta t) &\approx V(k) - c\Delta t V'(k) \\e^{-\rho\Delta t} &\approx 1 - \rho\Delta t\end{aligned}$$

Substitute into Bellman:

$$V(k) = \max_c \left\{ u(c)\Delta t + (1 - \rho\Delta t) [V(k) - c\Delta t V'(k)] \right\}$$

Expand:

$$V(k) = \max_c \left[u(c)\Delta t + V(k) - c\Delta t V'(k) - \rho\Delta t V(k) \right]$$

Cancel $V(k)$:

$$0 = \max_c [u(c)\Delta t - c\Delta t V'(k) - \rho\Delta t V(k)]$$

Continuous-Time Limit: HJB Equation (8–9)

Divide both sides by Δt :

$$0 = \max_c [u(c) - cV'(k) - \rho V(k)]$$

Rearrange:

$$\rho V(k) = \max_c \{u(c) - cV'(k)\}$$

Using $\dot{k} = -c$, we obtain the HJB:

$$\rho V(k) = \max_c \{u(c) + V'(k)\dot{k}\}$$

Key insight:

- Bellman: $V = u + \beta V'$
- HJB: $\rho V = u + V'(k)\dot{k}$

The Stable Arms

One can write the system as

$$\dot{z} = Az + \kappa, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Deviations near the steady state is

$$\Delta z = z - z^*$$

so

$$\dot{\Delta z} = A\Delta z$$

Characteristic equation

$$\det(A - \lambda I) = 0$$

yields the eigenvalues as solutions to:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Assume that there are two solutions $\lambda_1 > 0 > \lambda_2$

For each eigenvalue λ , the eigenvector $\mathbf{v}$ satisfies $(A - \lambda I)\mathbf{v} = 0$. Since $b \neq 0$, the convenient choice is

$$\mathbf{v}(\lambda) = \begin{pmatrix} b \\ \lambda - a \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{\lambda - a}{b} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{a - \lambda}{b} \end{pmatrix}$$

Then we can write Δz rigorously

$$\Delta z(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2, \quad \lambda_1 > 0, \lambda_2 < 0$$

Since $\lambda_1 > 0$, convergence to the steady state requires $c_1 = 0$. Set $c_1 = 0$, at $t = 0$, we have $\Delta z(0) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$ (here, c_1, c_2 are the coordinates of the initial deviation). Set explosive path $c_1 = 0$, we want to derive the value of y that "jumps" to the stable arm whenever there is deviation:

$$\Delta z = c_2 \mathbf{v}_2,$$

meaning

$$\Delta z = \begin{pmatrix} x - x^* \\ y - y^* \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{a - \lambda_2}{b} \end{pmatrix} \text{ or } y - y^* = -\frac{a - \lambda_2}{b}(x - x^*)$$

which is the policy function on the stable manifold

Further Readings

- Stokey, Nancy L., Robert E. Lucas, and Edward C. Prescott. Recursive Methods in Economic Dynamics. (1989). Harvard University Press.
- Essential mathematics for economic analysis (Sydsaeter & Hammond et al., 6e 2021)
- Further mathematics for economic analysis (Sydsaeter & Hammond et al., 2e 2008)
- Mathematics for economists (Simon & Blume, 1994)